

**Time:** 5 minutes**Outcome:** Measure Profit, Loss & Break-Even.**Difficulty:** ●●○ Intermediate

Profit, Loss & Break-Even



THE CORE IDEA

Choose the best operating output by comparing marginal revenue (MR) and marginal cost (MC). Then compare price (P) with average total cost (ATC): above ATC means profit, equal means break-even, and below means loss. Calculate total profit as $(P - ATC) \times Q$. A loss does not imply shutdown if revenue covers variable costs.



HOW TO RECOGNIZE IT

- Locate the operating output where MR crosses MC.
- Compare price with ATC for per-unit profit or loss; multiply by output for the total.
- A loss-making firm can still operate if revenue covers variable costs.

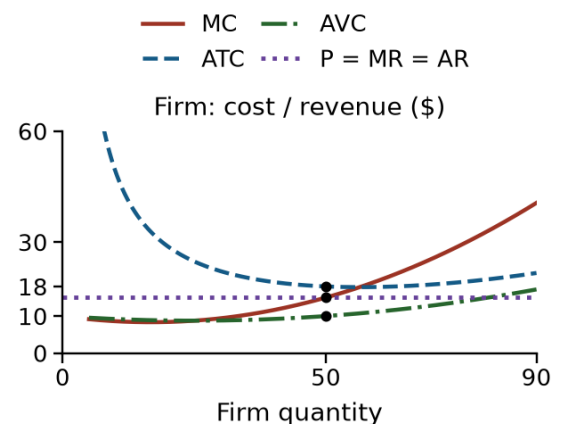
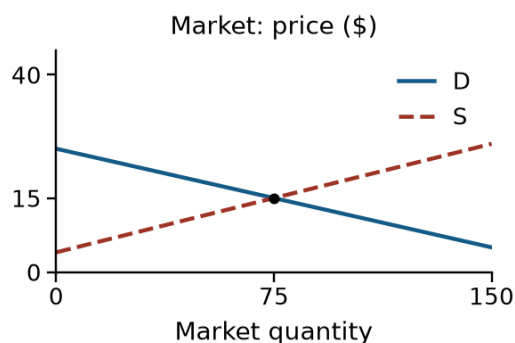


WATCH OUT

Do not confuse loss with shutdown. Profit/loss measurement and the short-run shutdown decision use different comparisons. Economic cost includes implicit opportunity costs, including the normal return needed to keep owner resources in the firm.



WORKED EXAMPLE: PROFIT, LOSS, OR BREAK-EVEN?



The market sets a price of \$15. The representative firm produces 50 units where $P = MR = MC$, but its ATC is \$18. Loss per unit is $\$18 - \$15 = \$3$, so total loss is $\$3 \times 50 = \150 . The firm may still operate because price exceeds AVC.



CHECK YOURSELF

How does the price-versus-ATC comparison connect a cost curve to the firm's profit status?

**READY?**

Return to the game and master this concept.